Product Owner case study · KYC onboarding

KYC approval fell 5.4 points in a month. Most of the arithmetic is known; most of the causes are not yet.

What the 2023 extracts show, what remains a hypothesis, and what to do first.

Data: KYC_summary and KYC_details extracts, 52,075 recorded decisions, Jul 25 – Sep 22, 2023 (prior period Jul 25 – Aug 23; recent period Aug 24 – Sep 22). Vendor research dated Oct 13, 2026. Case-study submission; it contains fraud-control and compliance findings from the extract.

The answer in seven points

1	What is measurable	Approval among recorded decisions, by day, document country and failed check. Not measurable: starts, completion, abandonment, retries, review outcomes, time to completion.
2	What changed	Pass rate 88.7% → 83.3% (−5.4 pp, 6.1% relative). Approvals 12.3% fewer (−2,945): 54% of that from fewer decisions, 46% from the lower rate.
3	Arithmetic drivers	Fraud signals −2.05 pp, execution errors −1.62 pp, selfie / liveness −0.95 pp, no document type −0.60 pp. Face match improved (+0.36 pp).
4	Still hypotheses	1,057 failures (range 1,055–1,130) sit above baseline in four date windows. Fraud signals are not adjudicated; the execution errors are unattributed; vendor incidents on Sep 1 and 6 are documented, but impact on ARQ is not.
5	First (Phase 0, weeks 0–2)	Instrument the funnel and hold and retry execution errors. Alert per reason code. Reconcile control findings and adjudicate fraud samples. Diagnose Mexican "no document type" with the vendor.
6	Needed	Event schema with privacy sign-off; vendor case access and incident reports; Risk and Compliance time; one medium and several small engineering items (sizes to confirm).
7	Verification	Acceptance criteria per action; experiments on eventual approval per person within 7 days. Scenarios for repeated causes: +0.9 pp to +4.3 pp, not forecasts. Confirmed control gaps could cost up to 2.3 pp.

We can measure approval among recorded decisions, not the start-to-finish funnel

Headline metric

Approval rate = approved (PASSED or APPROVED) ÷ recorded decisions.

Prior period 88.7%, recent 83.3%.

WARNING counts as not approved and not as a confirmed rejection. Its later review outcome is not in the data.

Each record is one user; the extract does not say whether it is a first, latest or final attempt, nor what event its date marks.

Metric	Status	Prior	Recent
Approval rate among recorded decisions	**Available**	88.7%	83.3%
Unresolved warning rate	**Available**	0.7%	1.1%
Completion rate (start to terminal state)	**Unavailable**	—	—
Eventual approval within 7 days	**Unavailable**	—	—
Abandonment rate	**Unavailable**	—	—
Retry rate and retry approval	**Unavailable**	—	—
Manual-review outcomes	**Unavailable**	—	—
Time to completion	**Unavailable**	—	—

To measure the funnel, log

- kyc_started
- document_selected
- document_captured
- selfie_captured
- kyc_submitted
- vendor_result
- kyc_decided
- review_completed
- account_enabled

with person, session and attempt ids, timestamps and app/SDK version.

Do not read the approval rate as completion. People who never submit, and approvals after retry or review, are not visible here. Whether true completion is lower or higher cannot be determined from these extracts.

Approval fell from 88.7% to 83.3%: 2,945 fewer approvals

88.7% → 83.3%

Approval among recorded decisions, prior → recent period (−5.4 pp)

−2,945

Approvals: −1,587 from fewer decisions at the prior rate, −1,358 from the lower rate

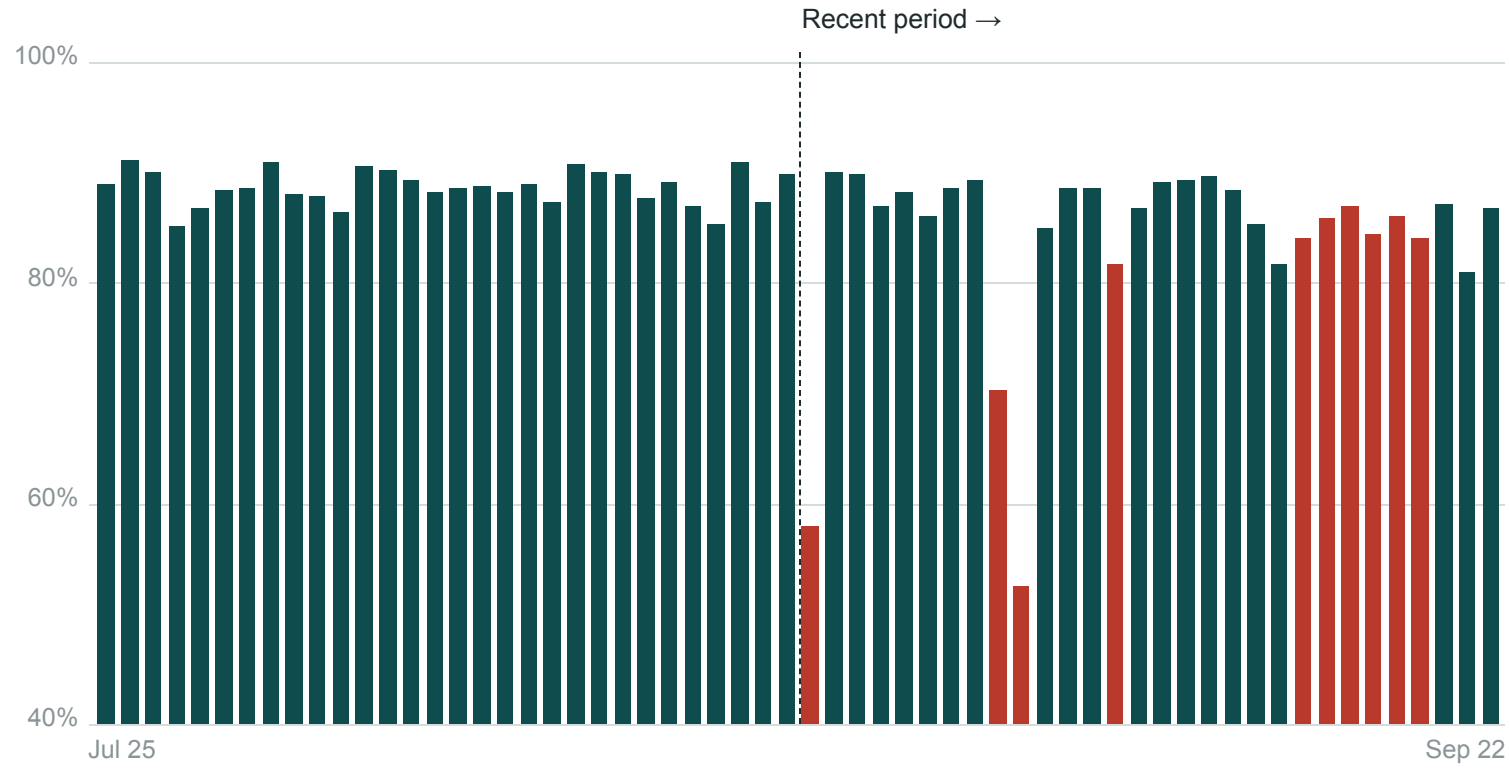
78.8%

Mexican documents (was 84.1%)

89.1%

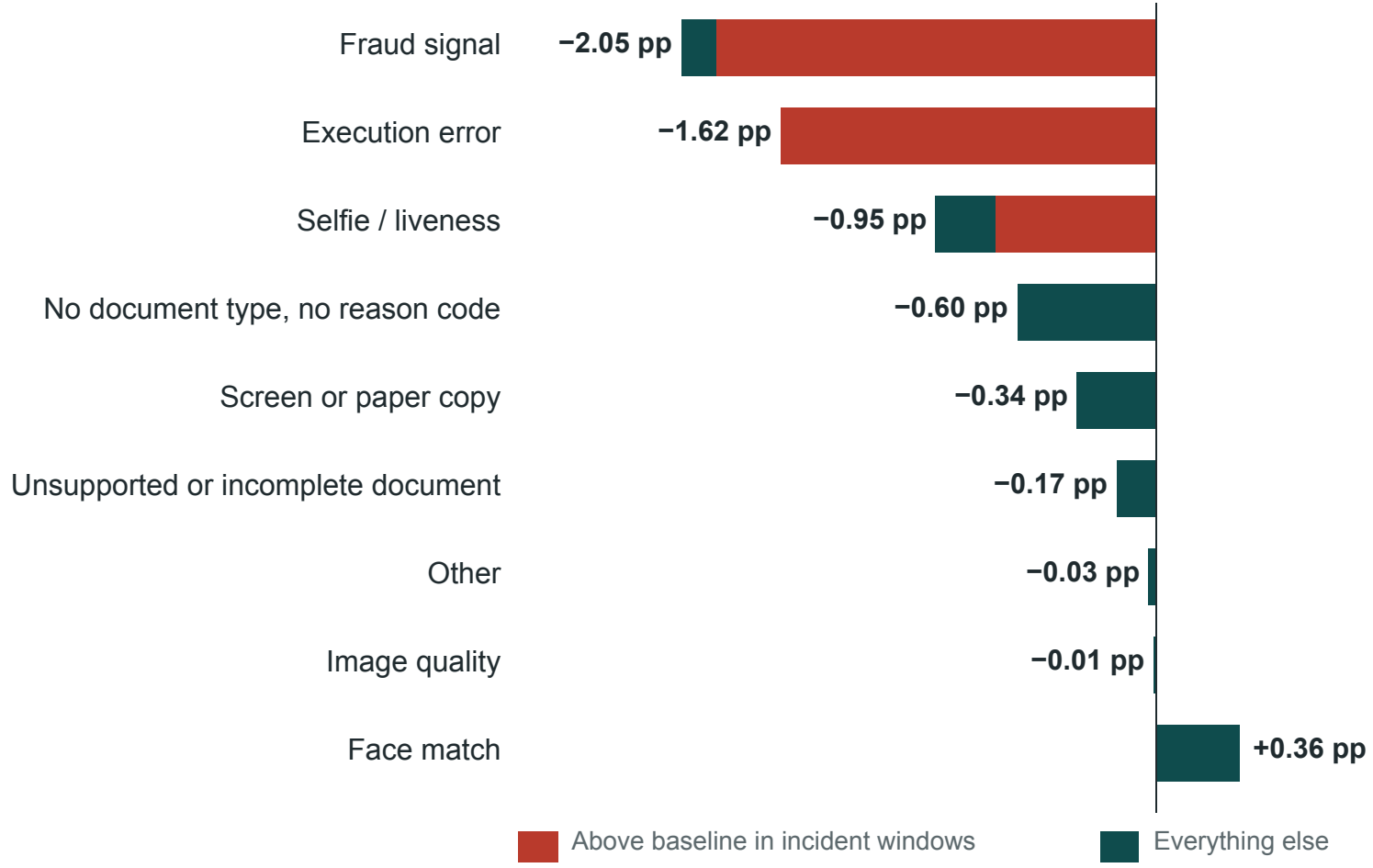
Argentine documents (was 94.9%)

Daily approval rate, all documents (red = incident window days)



Without the 10 incident days (red), the recent rate is 87.2%: −1.45 pp. Those days hold 38% of recent decisions, so they stay in. Mexican documents were 57.3% of decisions, then 56.8%. Columns start at 40%.

Four reasons explain most of the change; face match improved but remains a large loss



How to read the bars

- Fraud signals: −1.90 pp of −2.05 pp sits in two windows.
- Execution errors: almost all of −1.62 pp is on Sep 1–2.
- Selfie / liveness: −0.69 pp on Sep 1 and 6, −0.26 pp elsewhere.
- No document type, no reason code: −0.60 pp, all outside the windows, so it is a slower, continuous shift (mostly Mexican documents).
- Face match: 2.77% → 2.41% of decisions. Not a cause of the drop, but a large loss from before the drop.

298 recent unsuccessful decisions had two or more failed checks. A primary reason prevents double counts. A fix for that reason does not by itself get the person approved.

About 1,057 failures sit above baseline in four windows; causes are unconfirmed

Window (2023)	Type	Observed	Expected	Above baseline	Of recent unsuccessful	Approval pts
1. Argentina fraud-signal burst, Aug 24	Suspected additional attack traffic	376	2.9	373	8.9%	−1.48 pp
2. Execution errors at usability, Sep 1–2	Technical failure that affected applicants	407	0.1	407	9.7%	−1.62 pp
3. Liveness quality spike, Sep 1 and Sep 6	Technical failure that affected applicants	174	0.5	173	4.1%	−0.69 pp
4. Mexico INE fraud-signal cluster, Sep 14–19	Fraud-signal cluster without extra volume	111	7.0	104	2.5%	−0.41 pp
Total				**1,057**	**25.2%**	**−4.20 pp**

What we know and what we do not know

- Argentina, Aug 24: 660 decisions vs a typical 318; 373 of the flagged passed liveness. Suggests attack traffic; not adjudicated.
- Sep 1–2: usability TECHNICAL_ERROR, liveness passed in all 407, screening passed in 251. Vendor or ARQ integration: unknown.
- Mexico, Sep 14–19: INE flags without extra volume: fraud, a rule change or a new card version.

Vendor status page (read Oct 13, 2026)

Liveness incidents on the vendor's US data centre: Sep 1 (15:38–16:42 and 18:07–22:44 UTC) and Sep 6 (10:00–17:22 UTC). No entry for Sep 2 or for the fraud-signal dates.

A documented vendor incident is not confirmation of impact on ARQ.

Without the 10 incident days the rate still fell −1.45 pp (88.7% → 87.2%); they hold 38% of recent decisions. Sensitivity range 1,055–1,130 across window, signature and baseline variants. Without the 1,068 failures that match the patterns, the rate is 86.99%. That is a descriptive rate, not performance without incidents.

Six hypotheses, each with the step that would settle it

	Hypothesis	Strongest alternative	Next step	Decision it enables
H1	Execution errors on Sep 1–2 became rejections	Vendor fault vs ARQ integration or network	Pull ARQ logs and raw responses; vendor incident report	Fault owner; SLA question; hold-and-retry either way
H2	Liveness failures on Sep 1 and 6 were a vendor fault	ARQ app or SDK release	Release log; vendor list of affected transactions	Escalate; test a retake policy
H3	Aug 24 Argentina burst was coordinated fraud	Vendor false positives during a campaign	Adjudicate a sample; channel and device data	Pre-vendor velocity controls
H4	Sep 14–19 INE flags reflect a vendor rule change	Quieter fraud; new INE card version	Sample review with the vendor; registry check	Review vs auto-decline for INE flags
H5	Mexican "no document type" has several causes	Unsupported documents, capture, configuration, coverage	Review 100 cases with the vendor; config history	Picker vs configuration vs escalation
H6	Face mismatches partly reflect old ID photos	Impostors; poor light on the selfie	Adjudicate NO_MATCH and NOT_POSSIBLE samples	Review queue or registry face check

Data behind each: H1 407 TECHNICAL_ERROR rejections. H2 174 liveness quality failures on two days. H3 376 signals with doubled volume. H4 111 INE signals at normal volume. H5 no document type 11.3% → 14.2% of Mexican decisions. H6 face match 2.41% of recent decisions.

The plan in four tracks: fix and measure, test, learn, build

Track	P0 · Weeks 0–2 · Measure and stabilise	P1 · Weeks 2–6 · Test the main fixes	P2 · Weeks 6–13 · Scale what works	P3 · Next two quarters · Platform work (large)
Fix and measure Obvious fixes and the instruments to see them work. Cheap, start now.	• Minimum funnel instrumentation • Hold and retry on TECHNICAL_ERROR • Daily, then hourly, alerts per reason code • Reconcile the control findings • Report the headline with its breakdown	○ Velocity and device checks before the vendor call	○ Invite recoverable applicants back	
Test the fixes Hypotheses that are quick to try as experiments.		○ Test one guided retake on unclear selfies ○ Test accepted-document guidance in Mexico	○ On-device capture quality checks ○ Pilot manual review of face-match warnings	
Learn why Investigations and research that run beside the fixes.	• Adjudicate a sample of fraud signals • Diagnose 'no document type' in Mexico ○ Confirm vendor incident scope	○ Watch and ask people who failed the capture ○ Decide on the vendor: bake-off on a sample		
Build later Platform work that needs a quarter or more.				○ KYC orchestration and decision layer ○ Government registry checks (INE, RENAPER) ○ Second verification vendor for failover

○ = conditional on a Phase 0 finding. Owners, evidence, acceptance criteria and guardrails per action are in the dashboard (What to do). Effort sizes are assumptions.

Phase 0, weeks 0–2: measure and stabilise before we optimise

Minimum funnel instrumentation

Data & Analytics · Measure · M

Done when: Events for 99% of sessions in a daily reconciliation against vendor transactions; dashboards show first cohort.

Hold and retry on TECHNICAL_ERROR

Engineering · Onboarding platform · Remediate · S

Done when: Zero final rejections with TECHNICAL_ERROR over 4 weeks; 95% of held cases resolved within 24 hours.

Daily, then hourly, alerts per reason code

Data & Analytics · Measure · S

Done when: On replay, alerts fire on the first day of every incident window. False alerts in the prior period stay under one a week.

Reconcile the control findings

Compliance · Control · S

Done when: Every record classified (confirmed, explained, data error) with evidence.

Adjudicate a sample of fraud signals

Risk & Fraud · Investigate · S

Done when: At least 50 cases per cluster adjudicated with documented rationale.

Diagnose 'no document type' in Mexico

Product · Onboarding · Investigate · S

Done when: Each sampled case assigned a cause with vendor agreement.

Confirm vendor incident scope

Vendor management · Investigate · S

Done when: Written vendor confirmation or denial per date.

Report the headline with its breakdown

Data & Analytics · Measure · S

Done when: Weekly report adopted by the funnel review.

Effort sizes are assumptions (S ≤ 2 engineer-weeks, M = 2–6). Owners are role-based examples, not named people. Later phases: P1 weeks 2–6, P2 weeks 6–13, P3 the next two quarters.

Recovery scenarios: +0.9 pp to +4.3 pp on repeated causes, not a forecast

+0.9 pp

Conservative: 233 approvals per 30 days; recent rate would be 84.2%

+2.5 pp

Central: 634 approvals per 30 days; recent rate would be 85.8%

+4.3 pp

Optimistic: 1,093 approvals per 30 days; recent rate would be 87.6%

295

Central: approvals protected per comparable incident (execution errors, liveness spikes)

−580

If all control findings are confirmed: approvals per 30 days that we may need to reverse (−2.3 pp)

Recovered = observed population × eligibility × reach × return × remediation × subsequent approval

Population (recent period)	Observed	Central factors (E × R × Ret × Fix × Then)	Central	Range
Execution errors (Sep 1–2) (per incident)	407	1.0 × 1.0 × 0.75 × 0.9 × 0.88	242	116–351
Liveness quality failures on incident days (per incident)	174	0.75 × 1.0 × 0.6 × 0.75 × 0.9	53	23–84
Other selfie / liveness failures	486	0.66 × 1.0 × 0.6 × 0.45 × 0.9	78	28–138
Mexico: no document type or unsupported / incomplete document	1,570	0.7 × 1.0 × 1.0 × 0.35 × 0.84	323	107–561
Image quality and screen or paper copies	333	0.85 × 1.0 × 1.0 × 0.45 × 0.85	108	48–162
Face match: WARNING plus a share of NO_MATCH	278	1.0 × 1.0 × 1.0 × 0.45 × 1.0	125	50–232

A target set from these scenarios is a goal, not a prediction. "Eligible" for selfie populations is the observed share with no other failed check. "Then" is the prior-period approval rate for the same countries, except review, where a reviewer approval is final.

What we need, and how we will know it worked

Decisions and resources

- Approve Phase 0 and its owners.
- Privacy review of the event schema (Legal & Privacy).
- Vendor: case access, raw responses, incident reports for Sep 1, 2 and 6.
- Risk and Compliance time: fraud adjudication and control reconciliation.
- Engineering: instrumentation (M), hold-and-retry (S), alerts (S). Sizes to confirm.
- Finance: replace the value model’s market-based presets with ARQ unit economics.

How success is verified

- Zero final rejections with TECHNICAL_ERROR over 4 weeks.
- Alerts replayed on 2023 data fire on the first day of all four windows.
- Each control finding classified as confirmed, explained or data error.
- Experiments by person, outcome = eventual approval within 7 days. Mexico guidance: about 6,314 people per arm for +2 pp (stand-in baseline).
- Weekly: approval rate with fraud-signal, execution-error and warning shares beside it; completion once events exist.

Four things to confirm with the team before we act

	Question	Why it matters	What the answer changes
1	Does each row hold a user's first, latest or final KYC attempt, and does the date mark the submission or the vendor's decision?	The headline is approval among recorded decisions. If rows are latest attempts, we cannot see earlier failures and retries, and the first-pass rate is lower than shown. If rows are final decisions, the warnings can be out of date.	The metric definition (v2 to v3 in the registry), the retry hypotheses, and whether the bridge counts people or attempts.
2	Is there any record, even weekly counts, of people who opened KYC and never submitted?	The brief defines the pass rate as people who start and finish. Without starts, completion and abandonment cannot be measured, and the 1,587 fewer approvals from lower volume could be fewer starts, more abandonment, or both.	Whether Phase 0 instrumentation is a prerequisite or a check on current data; how we read the volume effect; the completion metric status.
3	What happens today after a TECHNICAL_ERROR or a liveness failure: does the user see the vendor's outcome, get a retake, or enter a review queue?	Hold-and-retry and the guided retake assume no retry exists. In the central scenario they recover 242 approvals per comparable execution-error incident and 78 per 30 days on steady liveness failures.	If a retry already exists, those recoverable counts shrink and the question becomes why retries fail; the Phase 0 hold-and-retry becomes a configuration check.
4	Which vendor product tier and data centre does ARQ use, and did ARQ receive incident notices for Sep 1, 2 and 6, 2023?	The vendor's status history shows liveness incidents on Sep 1 and 6 on its US data centre and nothing for Sep 2. The liveness spike is 173 failures above baseline and the execution errors 407.	On another data centre the liveness attribution loses its vendor evidence; with notices, the execution errors have an owner and an SLA claim.

The analysis stands on the extract alone. These answers decide how far each conclusion carries, not whether the work was done.

Appendix: evidence and method

A1 Definitions · A2 Validation · A3 Composition · A4 Mexico · A5 Incident sensitivity · A6 Experiments · A7–A8 Value model · A9 Operating model · A10 Control findings · A11 Vendor research

A1 · Metric definitions: what is available and what needs new events

Metric	Status	Definition and window	Gap or caveat
KYC starts	Unavailable	People who open the KYC flow. Window: daily, by country.	kyc_started with person id, session id, timestamp, country, app version, channel
Recorded decisions	Available	User records with a final decision in the extract (one per user_reference). Window: per period.	Whether a record is a first, latest or final attempt is not stated
Approval rate among recorded decisions	Available	Approved (PASSED or APPROVED) ÷ recorded decisions. The headline metric of this analysis. Window: per period, by decision date.	Not the start-to-finish completion rate: starts, abandonment and attempts are not in the data
Rejection rate	Available	REJECTED ÷ recorded decisions. Window: per period.	—
Unresolved warning rate	Available	WARNING ÷ recorded decisions. Later review outcome not in the extract. Window: per period.	Review dispositions needed
Completion rate (start to terminal state)	Unavailable	Starts that reach a terminal state (approved, finally rejected, or abandoned for 7 days) within 7 days of start. Window: 7 days after start.	kyc_started, step events, terminal decision, abandonment rule
Eventual approval within 7 days	Unavailable	Starts approved on any attempt within 7 days of the first start. Window: 7 days, cohort by start date.	Person id across attempts, attempt number, timestamps, review dispositions
Abandonment rate	Unavailable	Starts with no submission within 7 days. Window: 7 days.	kyc_started and kyc_submitted events
Retry rate and retry approval	Unavailable	Share of unsuccessful first attempts that retry, and their approval. Window: 7 days.	Attempt number per person
Manual-review outcomes	Unavailable	Approval and turnaround of cases sent to review. Window: per week.	Review queue events and dispositions
Time to completion	Unavailable	Median minutes from start to terminal decision. Window: per week.	Timestamps with time of day on start and decision
Execution-error rate	Available	Unsuccessful decisions whose primary reason is TECHNICAL_ERROR ÷ recorded decisions. Window: per period.	—
Fraud-signal rate	Available	Unsuccessful decisions whose primary reason is a manipulation or fake label ÷ recorded decisions. Not adjudicated. Window: per period.	Adjudication of a sample

A2 · Data validation and outcome treatment

Check	Result	Status	Treatment
Records in each extract	52,075 summary / 52,075 details	Pass	Equal row counts.
Unique user_reference	52,075 / 52,075	Pass	No duplicate users in either file.
Join coverage on user_reference	52,075 matched, 0 + 0 unmatched	Pass	Complete one-to-one join.
Date coverage	2023-07-25 to 2023-09-22: 60 of 60 calendar days	Pass	All days are present.
Timestamp precision	52,075 of 52,075 at midnight UTC	Limitation	Dates only. Intraday outage windows cannot be matched to records, and the extract does not say which event the date marks (submission, decision or export).
Final outcome categories (summary.decision_type)	PASSED 44,825, REJECTED 6,772, WARNING 474, APPROVED 4	Pass	PASSED and APPROVED = approved; REJECTED = rejected; WARNING = unresolved (not approved, not a confirmed rejection).
decision_type vs details.decision_label	PASSED vs REJECTED: 1; PASSED vs OK: 8	Note	summary.decision_type is treated as the outcome. The 8 'OK' labels are label-only; 1 record shows PASSED in the summary but REJECTED in details. We count it as approved until it is reconciled.
Non-standard capability value 'PASSES'	3 values	Note	Treated as PASSED.
Blank watchlist_screening_decision	6,606 blank, 1 NOT_EXECUTED	Note	Shown as 'not recorded'. Mostly on rejected records; a blank field does not prove screening never ran elsewhere.
Blank data_type (no document type returned)	3,781 records	Note	A symptom with several possible causes (unsupported document, capture, configuration, classification coverage).
Birth year extracted	48,286 present, 3,789 blank; range 1909–2022	Limitation	3,769 of the 3,789 blanks are rejected records, so ages describe applicants whose document was read, not all applicants. Extreme years (e.g. 2022) suggest extraction errors.
Country field	MEX / ARG only (data_issuing_country)	Note	This is the country that issued the document. It is not always the residence or the account market.
Attempts per user	1 record per user_reference	Limitation	The extract does not say whether a record is the first, latest or final attempt; retries and later outcomes are not visible.
Raw files unchanged	summary 67e4190909b7…, details e0762f377a41…	Pass	SHA-256 prefixes, recomputed on every build.

A3 · What the not-approved share is made of

Group	Class	Prior	Recent	Of decisions, prior	Of decisions, recent	Of unsuccessful, recent
Execution error	Execution error	1	407	0.00%	1.62%	9.7%
Fraud signal	Fraud signal (not adjudicated)	88	598	0.33%	2.38%	14.2%
No document type, no reason code	Unexplained or identity mismatch	945	1,034	3.51%	4.11%	24.6%
Unsupported or incomplete document	Capture or document issue	528	536	1.96%	2.13%	12.8%
Image quality	Capture or document issue	186	175	0.69%	0.70%	4.2%
Screen or paper copy	Capture or document issue	78	158	0.29%	0.63%	3.8%
Selfie / liveness	Capture or document issue	451	660	1.68%	2.63%	15.7%
Face match	Unexplained or identity mismatch	747	606	2.77%	2.41%	14.4%
Other	Unexplained or identity mismatch	21	27	0.08%	0.11%	0.6%
Fraud signal (not adjudicated)	Class total	88	598	0.33%	2.38%	14.2%
Execution error	Class total	1	407	0.00%	1.62%	9.7%
Capture or document issue	Class total	1,243	1,529	4.61%	6.08%	36.4%
Unexplained or identity mismatch	Class total	1,713	1,667	6.36%	6.63%	39.7%

Not-approved share: 11.3% prior, 16.7% recent. 281 recent decisions are unresolved warnings, not confirmed rejections.

A4 · Mexico: "no document type" is a symptom with several outcomes

Outcome for Mexican decisions with no document type	Prior	Recent	Share, prior	Share, recent	Change
No reason code	945	1,034	6.13%	7.24%	+1.11 pp
Unsupported document type	245	264	1.59%	1.85%	+0.26 pp
Mandatory data not present	203	204	1.32%	1.43%	+0.11 pp
Page or part not present	80	68	0.52%	0.48%	−0.04 pp
Image quality	186	175	1.21%	1.23%	+0.02 pp
Screen or paper copy	59	116	0.38%	0.81%	+0.43 pp
Execution error	1	96	0.01%	0.67%	+0.67 pp
Selfie, face or other	28	58	0.18%	0.41%	+0.22 pp
Approved	0	19	0.00%	0.13%	+0.13 pp

Document type returned (Mexico)	Approval prior	Approval recent
Voter ID (INE)	97.7%	93.9%
National ID card	91.0%	87.0%
No document type returned	0.0%	0.9%
Passport	97.8%	95.6%
Residence permit	97.6%	94.6%
Driving licence	97.0%	92.5%
Other	100.0%	93.8%

Argentine documents: 0 usability rejections or warnings in the extract; Mexican: 1,442. Possible causes include documents outside the accepted list, capture quality, a mismatch with the issuer country, configuration and vendor coverage.

A5 · Incident attribution: sensitivity and vendor evidence

Window	Base	Window ±1 day	Broader signature	Recent baseline	Old method (no baseline)
1. Argentina fraud-signal burst	373	378	372	371	376
2. Execution errors at usability	407	407	454	407	407
3. Liveness quality spike	173	174	171	174	174
4. Mexico INE fraud-signal cluster	104	102	133	103	111
Total	1,057				1,068

How to read it

- Base: failures that match the window's pattern, minus the count expected at the prior-period rate for the window's volume.
- Range across the four baseline-based variants: 1,055–1,130.
- The old method counts every failure of the pattern on the day. That includes failures that also occur on normal days.

Date	UTC	Vendor status page entry
2023-09-01	15:38–16:42	Liveness returned 'Unable to see your face properly' for all ID verification transactions.
2023-09-01	18:07–22:44	Same liveness message. Liveness restored 20:59; Authentication restored for new users only; resolved 22:44.
2023-09-06	10:00–17:22	Instability; workaround 11:25; liveness restored 16:21; resolved 17:22.
2023-09-13	13:23–13:44	Retry surge after a fix: V2 scans unavailable; Jumio Go customers possibly saw extra 'unsupported document' rejections; Hybrid delays.
2023-09-01 / 09-02	—	No status entry mentions usability, TECHNICAL_ERROR or skipped document checks, and none is listed for Sep 2.
2023-08-24, 09-14 to 09-19	—	No incidents listed for August 2023; only scheduled maintenance in Singapore during Sep 14–19.

A6 · Experiment and validation designs

	Mexico document guidance	Guided liveness retake
Kind	Optimisation experiment	Optimisation experiment
Hypothesis	Pictures of the accepted documents, shown to Mexican applicants before capture, increase eventual approval.	One guided retake after BAD_QUALITY or FACE_NOT_FULLY_VISIBLE from liveness increases eventual approval and does not increase fraud.
Population	People who start KYC with a Mexican document. Exclude staff and test accounts, and people already approved.	People whose first liveness result is BAD_QUALITY or FACE_NOT_FULLY_VISIBLE and who have no fraud signal. Exclude ID_USED_AS_SELFIE and any manipulation label.
Assignment unit	Person (person_id), assigned at first kyc_started, sticky across attempts.	Person, assigned at the first eligible liveness failure, sticky.
Primary outcome	Eventual approval within 7 days of first start.	Eventual approval within 7 days of the failure.
Window	7 days per person plus 7 days to mature the last cohort.	7 days plus 7 days to mature.
Minimum effect	+2.0 pp on eventual approval.	+10 pp on eventual approval in the eligible group.
Sample	With the recent Mexican approval rate (78.8%) as a stand-in baseline, the sample is about 6,314 people per arm (95% confidence, 80% power). At about 476 Mexican decisions a day that is roughly 27 days at a 50/50 split. Recompute once start-based data exists.	If about 5% of the eligible group is approved later without a retake (not measurable yet), the sample is about 141 people per arm. At about 16 eligible people a day (recent non-incident rate) that is roughly 18 days. Needs the real baseline from instrumentation.
Guardrails	Start-to-submit rate, time to decision, fraud-signal rate, support contacts.	Fraud-signal rate and confirmed fraud among retake approvals, liveness spoof labels, vendor cost per approval, time to decision.
Rules	Start at 10% for 3 days to check events, then 50/50. Expand if the primary outcome improves by at least the minimum and guardrails hold. Roll back if start-to-submit falls more than 2 pp or fraud signals rise more than 0.5 pp.	Needs Risk sign-off before launch. Start 50/50. Roll back on any confirmed spoof or a fraud-signal rate among retake approvals above control.

A7 · Value model inputs: market-based presets, not ARQ figures

Input	Low	Mid	High	Kind	Basis	Sources
Value per activated customer (USD)	$13	$21	$53	Derived	ARQ press figures; Nu México revenue and margin	S4, S5, S8, S9
Activation rate after approval (%)	38.2%	56.3%	83.0%	Proxy	Liftoff LATAM and fintech activation; Nu activity rate	S6, S7
Share of applicants from paid channels (%)	29.0%	40.0%	50.0%	Proxy + assumption	Nu organic and word-of-mouth shares	S6
Acquisition cost per paid applicant (USD)	$8.74	$17.96	$29.93	Derived	Liftoff cost per install and per registration	S7
Billable vendor transactions per decision (count)	1	1.2	1.5	Assumption	Assumption; replace with invoiced transactions ÷ decisions	S2
Vendor cost per billable transaction (USD)	$1.44	$1.85	$3	List price / estimate	Veriff and Sumsub list prices; Vendr estimate for Jumio	S1, S2, S3
Cost per manual review (USD)	$0.83	$5.72	$10.79	Derived	Mexico analyst salaries × time per case	S10, S11, S13, S14
Implementation cost of the repeated-cause actions (USD)	$24,400	$31,600	$85,000	Derived	Roadmap effort × Mexico engineering cost	S12, S13, S14
Run cost per month (USD)	$510	$2,010	$3,860	Derived	Review cases and retake vendor calls, central scenario	S1, S2, S3, S10, S11

Sources S1–S14: the dashboard (What we gain) gives each figure, period and link.

A8 · Value model results by preset, central scenario

Result (per 30 days unless stated)	Low	Mid	High	Cautious
Activated customers from recoverable approvals, per 30 days	242	357	526	242
First-year value of one 30-day cohort	$3,148	$7,496	$27,890	$3,148
Same, if every 30 days repeats for a year	$38,306	$91,199	$339,324	$38,306
Break-even first-year value per activated customer	$11	$13	$21	$45
Review cost per 30 days for the review-pilot population	$231	$1,590	$3,000	$3,000
Historical vendor spend on recent not-approved decisions	$6,049	$9,326	$18,905	$6,049
Vendor spend avoidable on suspected extra attack traffic	$492	$759	$1,539	$492
Historical paid acquisition spend on recent not-approved applicants	$10,648	$30,180	$62,868	$10,648
Pays back within the first year? (value per activated customer vs break-even)	Yes: $13 vs $11	Yes: $21 vs $13	Yes: $53 vs $21	No: $13 vs $45

The first-year value of a cohort comes in over the next year. It is not revenue recognised in the next 12 months. Past vendor and paid-acquisition spend on not-approved applicants is already spent. Only a fix of the cause stops future spend.

A9 · Operating model: owned metrics with versions, lineage and adoption

Metric	Status	Owner (role)	Version	Lineage
Approval rate among recorded decisions	Available	Product · Onboarding	2.0	KYC_summary. decision_type
Completion rate (start to terminal state)	Unavailable	Product · Onboarding	1.0	Needs new events
Eventual approval within 7 days	Unavailable	Product · Onboarding	1.0	Needs new events
Manual-review outcomes	Unavailable	Operations · KYC review	1.0	Needs new events
Execution-error rate	Available	Engineering · Onboarding platform	1.0	KYC_summary. decision_type
Fraud-signal rate	Available	Risk & Fraud	1.0	KYC_summary. decision_type

Ownership rules

- One accountable owner per metric, effective-dated.
- Definition changes need owner and Data approval; old versions stay queryable.
- Past results stay attributed to the owner at the time.

Adoption success measures

Share of KYC actions with an owner and a current status. Share of metrics with an approved definition. Time from alert to acknowledgement. Decisions in the review that cite the platform. Page views are not a success measure.

This analysis is version 2.0 of the headline metric: v1.0 called it "pass rate"; v2.0 renamed it and defined completion separately.

A10 · Control findings in the extract: to reconcile, not confirmed failures

Finding	Records	Other explanations	How to validate
Approved records with birth year 2006 or later	1,223	Extraction error; a policy that allows minors in some product; an approval record that never became an enabled account.	Compare with raw vendor responses and document images for a sample; check the age policy per country; check account state.
Approved records with no screening result in the extract	86	Screening ran in another system or a later step; field not exported.	Look up screening records for these users in the screening system; confirm whether an account was enabled.
Approved records with a watchlist WARNING	69	Warnings cleared in manual review that the extract does not show.	Retrieve review dispositions; confirm who cleared each case and when.
Outcome disagrees between the two files	9	A difference in the export field map.	Reconcile against the system of record.
Non-standard capability values	3	Manual edits or an export bug.	Fix at source; add schema checks to the extract.

If an extracted birth year of 2006 or later is accurate, the person was under 18 during 2023. The applicable eligibility policy and the data quality both still need confirmation.

A11 · Vendor research (Oct 13, 2026): availability is not effectiveness

Capability	2023 problem	Documented	Prerequisites	Not verified for ARQ
Classification	No document type returned	A separate capability that returns the document type, with UNSUPPORTED as one value.	None stated.	Whether ARQ's workflow already returns it.
Usability labels	Mandatory data not present; automation level	UNSUPPORTED_DOCUMENT_TYPE is a WARNING; MISSING_MANDATORY_DATAPOINTS 'will only happen for Jumio Go transactions'; TECHNICAL_ERROR is NOT_EXECUTED.	None stated.	Whether ARQ had manual review elsewhere in its own process.
Liveness labels and tiers	Liveness quality spike	BAD_QUALITY is a WARNING; FACE_NOT_FULLY_VISIBLE is REJECTED; Standard or Premium type.	Premium 'requires additional configuration for your tenant'; technology depends on the product purchased.	Which tier and labels ARQ receives.
Cross Transaction Risk	Fraud-signal clusters	Matches ID data, faces and image parts such as backgrounds across transactions; informational by default.	Jumio Support enables it on request. Press release of Apr 22, 2025 announced general availability.	Commercial terms; effect on ARQ's traffic.
Result retrieval and retries	Execution errors	After a callback, retry the retrieval from 40 seconds, with longer intervals each time, for at most 10 failures in a row.	None.	How ARQ handles TECHNICAL_ERROR today.
Mobile SDK capture features	No document type; capture quality	ML document and country classifier (4.9.0); unsupported-document check (4.11.0); virtual-camera injection detection (4.14.0); controls to limit manual capture (4.18.0).	SDK upgrade; configuration details for 4.18.0 controls not documented; gallery upload not documented.	ARQ's current SDK version and capture flow.
Mexico INE registry checks	INE fraud signals; face mismatch	INE Lista Nominal validates card fields. INE Data & Face Validation compares with INE records.	Data & Face needs a direct contract with INE, onboarding with FIMPE, dedicated middleware, about 4–5 days setup plus 2 days certification per client.	Cost, coverage and effect.
Argentina RENAPER data and biometric	Fraud signals; face mismatch	Data check and biometric (selfie vs registry) check.	PII fields need an updated data agreement (DPA) and are off by default; enabled by Jumio.	Cost and effect.

KYC approval-rate diagnosis: methodology and validation

Data: `KYC_summary.csv` and `KYC_details.csv`, 52,075 records each, Jul 25 – Sep 22, 2023. Vendor research dated 2026-10-13. Every figure below is read from `analysis/out/`, the same tables the dashboard and deck use.

1. What is measured

The brief defines the KYC pass rate as people who start and finish the process. The extracts hold one recorded decision per user, with a date and no time of day, and no record of starts, attempts or abandonment. The headline metric is therefore **approval among recorded decisions**. It is the number of decisions with a final outcome (`KYC_summary.decision_type`) of PASSED or APPROVED, divided by all recorded decisions in the period. It is not the start-to-finish completion rate, and the extracts cannot say whether that rate is higher or lower.

Metric	Status	Definition	Prior	Recent	Gap
KYC starts	Unavailable	People who open the KYC flow.	—	—	kyc_started with person id, session id, timestamp, country, app version, channel
Recorded decisions	Available	User records with a final decision in the extract (one per user_reference).	26,932	25,143	Whether a record is a first, latest or final attempt is not stated
Approval rate among recorded decisions	Available	Approved (PASSED or APPROVED) ÷ recorded decisions. The headline metric of this analysis.	88.69%	83.29%	Not the start-to-finish completion rate: starts, abandonment and attempts are not in the data
Rejection rate	Available	REJECTED ÷ recorded decisions.	10.59%	15.59%	—
Unresolved warning rate	Available	WARNING ÷ recorded decisions. Later review outcome not in the extract.	0.72%	1.12%	Review dispositions needed
Completion rate (start to terminal state)	Unavailable	Starts that reach a terminal state (approved, finally rejected, or abandoned for 7 days) within 7 days of start.	—	—	kyc_started, step events, terminal decision, abandonment rule
Eventual approval within 7 days	Unavailable	Starts approved on any attempt within 7 days of the first start.	—	—	Person id across attempts, attempt number, timestamps, review dispositions
Abandonment rate	Unavailable	Starts with no submission within 7 days.	—	—	kyc_started and kyc_submitted events
Retry rate and retry approval	Unavailable	Share of unsuccessful first attempts that retry, and their approval.	—	—	Attempt number per person
Manual-review outcomes	Unavailable	Approval and turnaround of cases sent to review.	—	—	Review queue events and dispositions

Metric	Status	Definition	Prior	Recent	Gap
Time to completion	Unavailable	Median minutes from start to terminal decision.	—	—	Timestamps with time of day on start and decision
Execution-error rate	Available	Unsuccessful decisions whose primary reason is TECHNICAL_ERROR ÷ recorded decisions.	0.00%	1.62%	—
Fraud-signal rate	Available	Unsuccessful decisions whose primary reason is a manipulation or fake label ÷ recorded decisions. Not adjudicated.	0.33%	2.38%	Adjudication of a sample

Outcome treatment. PASSED and APPROVED = approved; REJECTED = rejected; WARNING = unresolved (not approved, not a confirmed rejection); unsuccessful = rejected + warning. The 8 `OK` labels in the details file are label-only. The 1 record that is PASSED in the summary but REJECTED in details counts as approved until it is reconciled.

`PASSES` (3 values) is read as PASSED. Blank screening is "not recorded". NOT_EXECUTED with PRECONDITION_NOT_FULFILLED means an earlier check failed; with TECHNICAL_ERROR it is an execution error. Capability results are kept separate from the final outcome.

Periods. Prior period Jul 25 – Aug 23, 2023 and recent period Aug 24 – Sep 22, 2023: the two 30-day halves of the extract. Dates are those in `KYC_summary.date_`; which event they mark (submission, decision or export) is not stated.

2. Data validation

Check	Result	Status	Treatment
Records in each extract	52,075 summary / 52,075 details	Pass	Equal row counts.
Unique user_reference	52,075 / 52,075	Pass	No duplicate users in either file.
Join coverage on user_reference	52,075 matched, 0 + 0 unmatched	Pass	Complete one-to-one join.
Date coverage	2023-07-25 to 2023-09-22: 60 of 60 calendar days	Pass	All days are present.
Timestamp precision	52,075 of 52,075 at midnight UTC	Limitation	Dates only. Intraday outage windows cannot be matched to records, and the extract does not say which event the date marks (submission, decision or export).
Final outcome categories (summary.decision_type)	PASSED 44,825, REJECTED 6,772, WARNING 474, APPROVED 4	Pass	PASSED and APPROVED = approved; REJECTED = rejected; WARNING = unresolved (not approved, not a confirmed rejection).
decision_type vs details.decision_label	PASSED vs REJECTED: 1; PASSED vs OK: 8	Note	summary.decision_type is treated as the outcome. The 8 'OK' labels are label-only; 1 record shows PASSED in the summary but REJECTED in details. We count it as approved until it is reconciled.
Non-standard capability value 'PASSES'	3 values	Note	Treated as PASSED.

Check	Result	Status	Treatment
Blank watchlist_screening_decision	6,606 blank, 1 NOT_EXECUTED	Note	Shown as 'not recorded'. Mostly on rejected records; a blank field does not prove screening never ran elsewhere.
Blank data_type (no document type returned)	3,781 records	Note	A symptom with several possible causes (unsupported document, capture, configuration, classification coverage).
Birth year extracted	48,286 present, 3,789 blank; range 1909–2022	Limitation	3,769 of the 3,789 blanks are rejected records, so ages describe applicants whose document was read, not all applicants. Extreme years (e.g. 2022) suggest extraction errors.
Country field	MEX / ARG only (data_issuing_country)	Note	This is the country that issued the document. It is not always the residence or the account market.
Attempts per user	1 record per user_reference	Limitation	The extract does not say whether a record is the first, latest or final attempt; retries and later outcomes are not visible.
Raw files unchanged	summary 67e4190909b7…, details e0762f377a41…	Pass	SHA-256 prefixes, recomputed on every build.

Reconciliation checkpoints. Assertions in `build_datasets.py` reproduce each one:

- Prior period: 26,932 decisions, 23,887 approved (88.69%).
- Recent period: 25,143 decisions, 20,942 approved (83.29%).
- Change: −5.40 pp. Approved count −2,945: −1,587 from lower volume at the prior rate, −1,358 from the lower rate.
- Mexican documents 84.09% → 78.85%; Argentine 94.87% → 89.13%.

3. Attribution of unsuccessful decisions

Each unsuccessful decision gets one primary reason. TECHNICAL_ERROR in any capability comes first. Otherwise it is the first capability with REJECTED or WARNING, in vendor order: usability, image checks, extraction, data checks, liveness, face match. Then come a watchlist WARNING, "no document type, no reason code" and "no failed check". Reasons are grouped into nine groups and four classes (fraud signal, execution error, capture or document issue, unexplained or identity mismatch). Classes describe who could act on the vendor's reason; they are not adjudicated causes.

Because each decision has one primary reason, group effects sum to the total change. 298 recent unsuccessful decisions had two or more failed or warning checks, so a fix for the primary reason does not by itself mean approval.

Primary reason	Share prior	Share recent	Effect on approval	Above baseline in windows
Fraud signal	0.33%	2.38%	−2.05 pp	−1.90 pp
Execution error	0.00%	1.62%	−1.62 pp	−1.62 pp
Selfie / liveness	1.68%	2.62%	−0.95 pp	−0.69 pp
No document type, no reason code	3.51%	4.11%	−0.60 pp	+0.00 pp

Primary reason	Share prior	Share recent	Effect on approval	Above baseline in windows
Screen or paper copy	0.29%	0.63%	−0.34 pp	+0.00 pp
Unsupported or incomplete document	1.96%	2.13%	−0.17 pp	+0.00 pp
Other	0.08%	0.11%	−0.03 pp	+0.00 pp
Image quality	0.69%	0.70%	−0.01 pp	+0.00 pp
Face match	2.77%	2.41%	+0.36 pp	+0.00 pp

Face match improved between the periods. It is a large loss from before the drop, not a cause of the decrease.

4. Incident windows

A window is a date range, a country scope and a failure signature. Attributed failures = observed signature failures in the window − the count expected at the prior-period rate of that signature for the window's volume. Other failures on those days are not attributed. Records cannot be matched to intraday outages because all timestamps are midnight UTC.

Window (2023)	Type	Observed	Expected	Above baseline	Of recent decisions	Of recent unsuccessful	Approval points
Argentina fraud-signal burst, Aug 24	Suspected additional attack traffic	376	2.9	373	1.48%	8.9%	−1.48 pp
Execution errors at usability, Sep 1–2	Technical failure that affected applicants	407	0.1	407	1.62%	9.7%	−1.62 pp
Liveness quality spike, Sep 1 and Sep 6	Technical failure that affected applicants	174	0.5	173	0.69%	4.1%	−0.69 pp
Mexico INE fraud-signal cluster, Sep 14–19	Fraud-signal cluster without extra volume	111	7.0	104	0.41%	2.5%	−0.41 pp

Total above baseline 1,057 (25.2% of recent unsuccessful decisions, −4.20 pp of approval). Sensitivity across a ±1-day window, a broader signature and a recent-period baseline: 1,055–1,130. The earlier method (every failure of the pattern, no baseline) gives 1,068. Without the 1,068 selected failures, the recent rate is 86.99%. That is a descriptive rate, not a counterfactual.

Without the 10 incident days, the recent rate is 87.2% against 88.7% before: −1.45 pp. Those days hold 37.8% of recent decisions, so they are kept in the headline; the dashboard shows both views.

Vendor evidence. The vendor's public status history (monitor.jumio.com) was read on the research date. Source quality: primary vendor status page; some entries were read by a research sub-task and are marked as such.

Date	UTC	Entry	Read	ARQ impact
2023-09-01	15:38–16:42	Liveness returned 'Unable to see your face properly' for all ID verification transactions.	Fetched (agent)	Not confirmed
2023-09-01	18:07–22:44	Same liveness message. Liveness restored 20:59; Authentication restored for new users only; resolved 22:44.	Fetched	Not confirmed

Date	UTC	Entry	Read	ARQ impact
2023-09-06	10:00–17:22	Instability; workaround 11:25; liveness restored 16:21; resolved 17:22.	Fetched	Not confirmed
2023-09-13	13:23–13:44	Retry surge after a fix: V2 scans unavailable; Jumio Go customers possibly saw extra 'unsupported document' rejections; Hybrid delays.	Fetched (agent)	Not confirmed
2023-09-01 / 09-02	—	No status entry mentions usability, TECHNICAL_ERROR or skipped document checks, and none is listed for Sep 2.	Fetched (agent)	Unknown: vendor or integration
2023-08-24, 09-14 to 09-19	—	No incidents listed for August 2023; only scheduled maintenance in Singapore during Sep 14–19.	Fetched (agent)	Not applicable

A documented vendor incident is not confirmation of impact on ARQ: ARQ's data centre, product tier and transaction times are unknown. No vendor entry matches the Sep 1–2 TECHNICAL_ERROR pattern, so those errors could be the vendor's or ARQ's integration.

5. Recovery scenarios

Recovered approvals = observed population × eligibility × reach × return × remediation × subsequent approval. Observed populations are recent-period counts by primary reason and do not overlap. All other factors are assumptions, shown per scenario. For the selfie populations, eligibility is the observed share with no other failed check.

Subsequent approval uses the prior-period approval rate for the same countries, because checks that did not run could still fail. The review queue is the exception: a reviewer approval is final. Incident-type benefits are reported per comparable incident, not per month. Possible approval reversals from confirmed control findings are reported separately.

Scenario	Recoverable per 30 days	Approval points	Recent rate with them	Per comparable incident
Conservative	233	+0.9 pp	84.2%	139
Central	634	+2.5 pp	85.8%	295
Optimistic	1,093	+4.3 pp	87.6%	435

If all control findings were confirmed, up to 580 recent approvals (−2.3 pp) we might need to reverse. These are scenarios built on stated assumptions, not forecasts; a target set from them is a goal, not a prediction.

6. Experiment sample sizes

Per-arm sample sizes use the normal approximation for two proportions (two-sided $\alpha = 0.05$, power 0.80). Assignment is per person, so repeat attempts do not count as independent users.

Mexico document guidance. With the recent Mexican approval rate (78.8%) as a stand-in baseline, the sample is about 6,314 people per arm (95% confidence, 80% power). At about 476 Mexican decisions a day that is roughly 27 days at a 50/50 split. Recompute once start-based data exists.

Liveness retake. If about 5% of the eligible group is approved later without a retake (not measurable yet), the sample is about 141 people per arm. At about 16 eligible people a day (recent non-incident rate) that is roughly 18 days. Needs the real baseline from instrumentation.

Hold-and-retry for TECHNICAL_ERROR is remediation of a confirmed failure and is validated by acceptance criteria, not randomised.

7. Value model

The model keeps two kinds of numbers apart. Observed quantities come from the data: the approved-count change, the rate counterfactual, recoverable approvals per scenario and suspected extra attack traffic. Financial inputs come from outside the data: value and activation per customer, paid-channel share and cost, vendor cost, review cost, and build and run cost.

The inputs start from market-based presets derived in `analysis/value_presets.py` from published sources fetched on the research date. None is an ARQ figure; each should be replaced with ARQ's own number.

Low, Mid and High are the low, middle and high values of each range, so a High preset raises costs as well as value. Cautious takes the weaker end of each range. Results appear only when every required input is valid. The annual figure multiplies 30-day figures by 365/30. First-year cohort value is not revenue recognised in the next 12 months.

Input	Low	Mid	High	Kind	Basis	Sources
Value per activated customer (USD)	$13	$21	$53	Derived	Low: ARQ's reported US$10bn volume ÷ 2m users × 0.5% fee = $25 revenue a year, less $12 cost to serve (Nu México, $1 a month). Mid: Nu México ARPAC $12.50 × 12 × 40% first-year ramp (assumption) × 35.3% gross margin. High: the same without the ramp, an average Nu México customer. Not all reported volume carries the fee, and users are not all active.	S4 S5 S8 S9
Activation rate after approval (%)	38.2%	56.3%	83.0%	Proxy	Low: Liftoff LATAM activation rate 38.2%. Mid: Liftoff fintech activation rate 56.3% (2021–22 data; the source does not define activation). High: Nu's monthly activity rate of 83%, a ceiling set by a mature primary bank. None measures approved-to-active directly.	S6 S7
Share of applicants from paid channels (%)	29.0%	40.0%	50.0%	Proxy + assumption	Low: 29% of Nu's new Brazilian customers in 2025 were not sourced organically. High: Nu acquired more than 50% of Mexican customers through referrals, so other channels are under 50%. Mid: 40% (assumption). Not every non-organic channel is paid.	S6
Acquisition cost per paid applicant (USD)	$8.74	$17.96	$29.93	Derived	Low: Liftoff LATAM cost per install $1.60 ÷ 18.3% registration rate = $8.74 per registration. Mid: Liftoff fintech cost per registration $17.96. High: $17.96 ÷ 60% of registrants who reach KYC (assumption). 2021–22 data.	S7
Billable vendor transactions per decision (count)	1	1.2	1.5	Assumption	Low: one billable transaction per decision. Mid and high: 1.2 and 1.5 with retries (assumption). Some vendors (Sumsub) charge only for successful verifications, which would make not-approved decisions free.	S2

Input	Low	Mid	High	Kind	Basis	Sources
Vendor cost per billable transaction (USD)	$1.44	$1.85	$3	List price / estimate	Low: Veriff Essential $0.80 + PEP and sanctions $0.64. Mid: Sumsub Compliance $1.85, which includes AML screening. High: top of Vendr's $1.50–$3.00 estimate for Jumio at 100,000–500,000 verifications a year; the extract annualises to about 316,790 decisions. Self-serve list prices; negotiated enterprise rates are usually lower.	S1 S2 S3
Cost per manual review (USD)	$0.83	$5.72	$10.79	Derived	Monthly gross salary × loaded factor ÷ MXN 18.3528 ÷ 140 productive hours × minutes per case × quality uplift. Low: MXN 11,236 (Paylab, lowest 10% of financial analysts) × 1.3, 8 min, ×1.1. Mid: MXN 39,608 (TalentUp KYC analyst median) × 1.35, 15 min, ×1.1. High: the same salary × 1.4, 25 min, ×1.2. Hours, minutes and uplift are assumptions.	S10 S11 S13 S14
Implementation cost of the repeated-cause actions (USD)	$24,400	$31,600	$85,000	Derived	Engineer-weeks × loaded weekly cost × team overhead. Weeks: 12 / 13 / 19.5, from the roadmap's effort for the four 30-day actions plus 2 weeks for review tools; high adds a 50% overrun. Weekly cost: Levels.fyi median MX$1,146,490 ÷ 18.3528 × 1.3 or × 1.35 ÷ 52 ($1,562, $1,622); high uses Revelo's top fully loaded $126,000 ($2,423). Overhead for product, design, QA and vendor work: ×1.3 / ×1.5 / ×1.8 (assumption).	S12 S13 S14
Run cost per month (USD)	$510	$2,010	$3,860	Derived	Central scenario, per 30 days: 278 review-pilot cases × cost per review. Add about 191 guided selfie retakes × billable transactions × cost per transaction, with the same preset's values. Excludes support contacts and alert tools.	S1 S2 S3 S10 S11

Sources (fetched on the research date; status shows how each was obtained):

ID	Source	Figure used	Period	Status
S1	Veriff: Pricing page, self-serve plans	Essential $0.80, Plus $1.39, Premium $1.89 per verification; PEP and sanctions add-on +$0.64	List prices, 13 Oct 2026	Fetched
S2	Sumsub: Pricing and plans	Basic $1.35 (ID, liveness, face match); Compliance $1.85 (adds AML screening) per verification; only successful verifications are charged	List prices, 13 Oct 2026	Fetched
S3	Vendr: Jumio buyer guide	Estimated $1.50–$3.00 per verification at 100,000–500,000 a year; median contract $55,850 a year	Updated Feb 2026	Fetched (third-party estimate)
S4	Nu Holdings: Nu México press release	Nu México ARPAC US$12.5 a month; cost to serve US$1 a month	Q3 2025, published 11 Feb 2026	Fetched
S5	Nu Holdings: Form 6-K, Q1 2026 results	Gross profit US$1,877.7m on revenue US$5,315.5m (managerial P&L), a 35.3% margin (derived)	Q1 2026	Fetched
S6	Nu Holdings: Form 20-F, FY2025	Monthly activity rate 83%; 71% of new Brazilian customers in 2025 sourced organically; more than 50% of Mexican customers acquired through word-of-mouth referrals	31 Dec 2025	Fetched

ID	Source	Figure used	Period	Status
S7	Liftoff, via MarTech Series: Mobile Finance Apps report	Fintech cost per registration $17.96 and activation rate 56.3%; LATAM cost per install $1.60, registration rate 18.3%, activation rate 38.2%	Data Jun 2021 – Jun 2022	Fetched (secondary, dated)
S8	Bloomberg Línea: DolarApp raises US$70m and becomes ARQ	More than 2 million users; more than US$10bn annualised transaction volume	3 Mar 2026	Fetched (press)
S9	Passageiro de Primeira: ARQ keeps a 0.5% fee on any conversion	0.5% of the converted amount, no minimum	28 Jul 2026	Fetched (consumer site)
S10	TalentUp: KYC analyst salary, Mexico City	Median gross MXN 475,300 a year	Not stated (2026 page)	Fetched
S11	Paylab: Financial analyst salary, banking, Mexico	Monthly gross: the lowest 10% earn under MXN 11,236 and the highest 10% over MXN 38,290; fewer than 20 verified respondents	Not stated	Fetched
S12	Levels.fyi: Senior software engineer, Mexico	Median total compensation MX$1,146,490 a year (224 submissions)	Updated 13 Oct 2026	Fetched (self-reported)
S13	Revelo: Hiring software developers in Mexico	Senior $66,000–$90,000 a year; fully loaded cost 1.3–1.4× salary (IMSS, Infonavit, aguinaldo, PTU), up to $126,000	3 Jul 2026	Fetched (recruiter)
S14	Banco de México: FIX exchange rate	MXN 18.3528 per USD	13 Oct 2026	Fetched

Results in the central scenario (634 recoverable approvals per 30 days):

Result	Low	Mid	High	Cautious
Activated customers from recoverable approvals, per 30 days	242	357	526	242
First-year value of one 30-day cohort	$3,148	$7,496	$27,890	$3,148
Same, if every 30 days repeats for a year	$38,306	$91,199	$339,324	$38,306
Break-even first-year value per activated customer	$11	$13	$21	$45
Review cost per 30 days for the review-pilot population	$231	$1,590	$3,000	$3,000
Historical vendor spend on recent not-approved decisions	$6,049	$9,326	$18,905	$6,049
Vendor spend avoidable on suspected extra attack traffic	$492	$759	$1,539	$492
Historical paid acquisition spend on recent not-approved applicants	$10,648	$30,180	$62,868	$10,648

Under the Cautious preset the actions do not pay back within the first year: break-even is $45 per activated customer against $13 of value. That is the case Finance should test first with ARQ's own figures.

8. Checks performed for this submission

- `build_datasets.py` asserts the reconciliation checkpoints. It also asserts that class shares sum to the not-approved share and that process-step counts reconcile to all unsuccessful decisions.
- Each action has exactly one accountable role (asserted). Each run records the SHA-256 hashes of both raw files.

- Every number marked in the dashboard page resolves to a row in `analysis/out/` (checked by script); every table column exists in its dataset.
- The export embeds no `user_reference` values and no record-level columns (asserted in `export_html.py`), loads no network resources, and bundles d3 7.9.0.
- Browser tests of the export: one tab and one audience panel show at a time. Tooltips hide when the pointer leaves. The country filter updates its three views.
- More browser tests: sort and search work; the scenario picker updates tiles, table and value model; each preset gives the results in `value_results.csv`.
- Invalid and blank inputs show "Needs …" and no old values. Keyboard tab navigation works. Nothing overflows at 375 px; the console has no errors.
- Print to PDF shows all tabs, details and charts.
- On a replay of the extract, the alert rules fire on the first day of all four windows. 5 rule-days fire in the prior period, so the thresholds need adjustment.
- The PowerPoint passes the pptx skill's structural validator; its text was checked with markitdown; every slide was rendered from the shared slide specification and inspected.

9. Open questions for ARQ

These are not asked in the submission. They are the answers that would change the findings, in the order they matter; each says what changes with the answer.

#	Question	Why it matters	What the answer changes	Who would know
1	Does each row hold a user's first, latest or final KYC attempt, and does the date mark the submission or the vendor's decision?	The headline is approval among recorded decisions. If rows are latest attempts, we cannot see earlier failures and retries, and the first-pass rate is lower than shown. If rows are final decisions, the warnings can be out of date.	The metric definition (v2 to v3 in the registry), the retry hypotheses, and whether the bridge counts people or attempts.	Data platform or whoever produced the extract.
2	Is there any record, even weekly counts, of people who opened KYC and never submitted?	The brief defines the pass rate as people who start and finish. Without starts, completion and abandonment cannot be measured, and the 1,587 fewer approvals from lower volume could be fewer starts, more abandonment, or both.	Whether Phase 0 instrumentation is a prerequisite or a check on current data; how we read the volume effect; the completion metric status.	Mobile and platform engineering (app analytics, API logs).
3	What happens today after a TECHNICAL_ERROR or a liveness failure: does the user see the vendor's outcome, get a retake, or enter a review queue?	Hold-and-retry and the guided retake assume no retry exists. In the central scenario they recover 242 approvals per comparable execution-error incident and 78 per 30 days on steady liveness failures.	If a retry already exists, those recoverable counts shrink and the question becomes why retries fail; the Phase 0 hold-and-retry becomes a configuration check.	Product and the onboarding squad; the vendor integration owner.

#	Question	Why it matters	What the answer changes	Who would know
4	Which vendor product tier and data centre does ARQ use, and did ARQ receive incident notices for Sep 1, 2 and 6, 2023?	The vendor's status history shows liveness incidents on Sep 1 and 6 on its US data centre and nothing for Sep 2. The liveness spike is 173 failures above baseline and the execution errors 407.	On another data centre the liveness attribution loses its vendor evidence; with notices, the execution errors have an owner and an SLA claim.	Vendor management; the vendor's account team.

10. Limitations and unresolved dependencies

- Starts, abandonment, attempts, review outcomes, account state and times of day are not in the extracts. Completion and eventual approval stay unavailable until the Phase 0 events exist.
- Whether a record is a first, latest or final attempt, and what event its date marks, is unknown; recent cohorts may be immature.
- Fraud signals are not adjudicated; execution errors are not attributed; vendor incident impact on ARQ is unconfirmed.
- Birth year is extracted data with outcome-related missingness; ages describe people whose document was read.
- Country is the country that issued the document.
- ARQ's current flow (retries, review, upload policy, architecture, vendor product tier) is unknown; the target design and roadmap mark these as discovery questions or conditions.
- Effort sizes, scenario factors and the role-based organisation are assumptions. The value-model presets come from public prices, salaries and benchmarks for other companies (several dated or third-party estimates), not from ARQ's contracts or finance data.
- Chrome rendered the deck PDF and the visual QA images from the HTML version of the slide specification. No PowerPoint or LibreOffice renderer was available.
- The .pptx passes a structural validator, but nobody opened it in PowerPoint, so table row heights can be slightly different there.

11. Reproduce

```
bash analysis/build_all.sh
```

This runs `build_datasets.py`, `export_html.py`, `build_notes.py` and `tools/build_deck.mjs`, then prints the deck PDF with headless Chrome. The tools in `analysis/tools/package.json` are pinned (d3 7.9.0, pptxgenjs 3.12.0).